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Subject: Programming with Java	**Semester: III**
ASSIGNMENT No : 21	**Year: AY 2026-27**
TITLE : JDBC DriverManager, Connection, and Statement	

Problem Statement

Demonstration and exploration of JDBC classes and interfaces (DriverManager, Connection, Statement).

Learning Objectives

To understand and use the core JDBC classes for database connectivity.

Program:

student_db.sql

```sql
CREATE DATABASE student_db;
USE student_db;

CREATE TABLE connection_test (
   test_id INT PRIMARY KEY,
   status VARCHAR(50)
);

INSERT INTO connection_test VALUES (1, 'Database Operational');
```

DBConnectionTest.java

```java
// Assignment 21 - Exercise 1: Connect Java with Database and Display
Status
import java.sql.*;
public class DBConnectionTest
{
   public static void main(String args[]) // Main Method
   {
       String url = "jdbc:mysql://localhost:3306/student_db"; //
Database URL
       String user = "thegharkalaptop"; // DB Username
       String pass = ""; // DB Password
```

```
        try // Try block
        {
            System.out.println("Loading MySQL JDBC Driver..."); // Driver
loading message
            Class.forName("com.mysql.cj.jdbc.Driver"); // Loading driver
dynamically
            System.out.println("Driver loaded successfully!");

            System.out.println("Attempting connection to database
server..."); // Attempt message
            Connection con = DriverManager.getConnection(url, user,
pass); // Establishing connection using DriverManager

            if(con != null && !con.isClosed()) // Verifying connection
status
            {
                System.out.println("\n[SUCCESS] Database Connection
Status: CONNECTED"); // Status confirmation
                System.out.println("Connected to: " +
con.getMetaData().getDatabaseProductName());
                System.out.println("Driver Version: " +
con.getMetaData().getDriverVersion()); // Displaying driver version
            }
            else // Closed condition
            {
                System.out.println("\n[FAILED] Database Connection
Status: NOT CONNECTED"); // Failure status
            }

            con.close(); // Closing connection session
            System.out.println("Database connection closed cleanly.");
        }
        catch(ClassNotFoundException e) // Catching missing driver jar
        {
            System.out.println("Driver Error: MySQL JDBC Driver not found
in classpath!"); // Driver error
        }
        catch(SQLException e) // Catching connection/credential errors
        {
            System.out.println("SQL Connection Error: " +
e.getMessage()); // SQL error
        }
    }
```

```
}
```

StudentDBConnection.java

```
// Assignment 21 - Exercise 2: Connect to Student Database and Display
Confirmation Message
import java.sql.*;

public class StudentDBConnection
{
   public static void main(String args[]) // Main Method
   {
       String url = "jdbc:mysql://localhost:3306/student_db"; // URL
targeting student_db
       String user = "thegharkalaptop"; // DB Username
       String pass = ""; // DB Password

       try // Exception handling block
       {
           Class.forName("com.mysql.cj.jdbc.Driver"); // Registering
driver
           Connection con = DriverManager.getConnection(url, user,
pass); // Connecting via DriverManager

           if(con != null && !con.isClosed()) // Validating connection
           {
               String dbName = con.getCatalog(); // Fetching current
database name from connection session
               System.out.println("Connection Status: ACTIVE");
               System.out.println("Message: The " + dbName + " database
is connected successfully!");

               Statement stmt = con.createStatement(); // Creating
Statement to explore interface
               ResultSet rs = stmt.executeQuery("SELECT DATABASE()"); //
Querying active DB catalog
               if(rs.next())
               {
                   System.out.println("Verified Active Catalog from
Server: " + rs.getString(1));
               }

               rs.close(); // Closing ResultSet
               stmt.close(); // Closing Statement
```

```
            }
            con.close(); // Closing Connection
        }
        catch(Exception e) // Catching general connection exceptions
        {
            System.out.println("Connection Failed: " + e.getMessage());
// Error message
        }
    }
}
```

Output:

DBConnectionTest.java

```
Problems  Output  Debug Console  Terminal  Ports

● thegharkalaptop@The-Ghar-Ka-Laptop JavaSQL % javac DBConnectionTest.java && java -cp ".:lib/*" DBConnectionTest.java
  Loading MySQL JDBC Driver...
  Driver loaded successfully!
  Attempting connection to database server...

  [SUCCESS] Database Connection Status: CONNECTED
  Connected to: MySQL
  Driver Version: mysql-connector-j-26.7.0 (Revision: 559f62df01d4f618440da174ff45ba87f7b3b2a8)
  Database connection closed cleanly.
○ thegharkalaptop@The-Ghar-Ka-Laptop JavaSQL %
```

```
Problems  Output  Debug Console  Terminal  Ports

  cd "/Users/thegharkalaptop/Documents/JavaSQL/" && javac DBConnectionTest.java && java DBConnectionTest
● thegharkalaptop@The-Ghar-Ka-Laptop JavaSQL % cd "/Users/thegharkalaptop/Documents/JavaSQL/" && javac DBConnectionTest.java && java DBConnection
  Test
  Loading MySQL JDBC Driver...
  Driver Error: MySQL JDBC Driver not found in classpath!
○ thegharkalaptop@The-Ghar-Ka-Laptop JavaSQL %
```

StudentDBConnection.java

```
Problems  Output  Debug Console  Terminal  Ports

● thegharkalaptop@The-Ghar-Ka-Laptop JavaSQL % javac StudentDBConnection.java && java -cp ".:lib/*" StudentDBConnection.java
  Connection Status: ACTIVE
  Message: The student_db database is connected successfully!
  Verified Active Catalog from Server: student_db
○ thegharkalaptop@The-Ghar-Ka-Laptop JavaSQL %
```

```
Problems  Output  Debug Console  Terminal  Ports

  cd "/Users/thegharkalaptop/Documents/JavaSQL/" && javac StudentDBConnection.java && java StudentDBConnection
● thegharkalaptop@The-Ghar-Ka-Laptop JavaSQL % cd "/Users/thegharkalaptop/Documents/JavaSQL/" && javac StudentDBConnection.java && java StudentDB
  Connection
  Connection Failed: com.mysql.cj.jdbc.Driver
○ thegharkalaptop@The-Ghar-Ka-Laptop JavaSQL %
```

Conclusion:

In this assignment, the foundational JDBC classes and interfaces, DriverManager, Connection, and Statement, were explored and verified for establishing reliable database communications. The DriverManager successfully registered the MySQL driver and managed authentication sessions, the Connection interface maintained an active bridge to the student_db schema, and the Statement interface executed metadata queries to evaluate system readiness, demonstrating the core mechanics of Java database connectivity.